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Cryptography Network and Security

1) Given:

Plaintext : MEETATNOON.

Numerical value : A=0, B=1 ... Z=25

Formula : $C = (P+7) \bmod 26$.

$P = (C-7) \bmod 26$.

Repeating key : sun.

Solu: Encryption formula : $C = (P+7) \bmod 26$.

Decryption formula : $P = (C-7) \bmod 26$.

A=0 B=1 C=2 D=3 E=4 F=5 G=6 H=7 I=8

J=9 K=10 L=11 M=12 N=13 O=14 P=15 Q=16 R=17

S=18 T=19 U=20 V=21 W=22 X=23 Y=24 Z=25.

1) Encryption :

Plain text: M E E T A T N O O N.

12 4 4 19 0 19 13 14 14 13.

Taking mod 26 : Sub value in formula.

M (12)	$C = (12+7) \bmod 26$	19. (T)
E (4)	$C = (4+7) \bmod 26$.	11 (L)
E (4)	$C = (4+7) \bmod 26$	11 (L)
T (19)	$C = (19+7) \bmod 26$	1 (B)
A (0)	$C = (0+7) \bmod 26$	7 (H)
19 (T)	$C = (19+7) \bmod 26$	1 (B)

N (13) C = (13+7) mod 26. 20(U)

O (14) C = (14+7) mod 26. 21(V)

O (14) C = (14+7) mod 26. 21(V)

N (13) C = (13+7) mod 26 20(U).

∴ Ciphertext = TLLB HB UVVU.

Decryption:

Using the formula : P = (C-7) mod 26.

	T	L	L	B	H	B	U	V	V	U
	19	11	11	1	7	1	20	21	21	20.

T(19) P = (19-7) mod 26 12(M).

L(11) P = (11-7) mod 26 4(E)

L(11) P = (11-7) mod 26 4(E).

B(1) P = (1-7) mod 26 19(T).

H(7) P = (7-7) mod 26 0(A).

B(1) P = (1-7) mod 26. 19(T).

U(20) P = (20-7) mod 26 13(N)

V(21) P = (21-7) mod 26 14(O)

V(21) P = (21-7) mod 26. 14(O).

U(20) P = (20-7) mod 26. 13(N).

∴ The Decrypted message : MEETATNOON.

Repeating key : Sun :

Plaintext : MEETATNOON.

Key : Sun

M	E	E	T	A	T	N	O	O	N
(12)	(4)	(4)	(19)	(0)	(19)	(13)	(14)	(14)	(13)
S	U	N	S	U	N	S	U	N	S
(18)	(20)	(13)	(18)	(20)	(13)	(18)	(20)	(13)	(18)

(i) Encryption formula : $C = (P+K) \bmod 26$.

M (12) $C = (12+18) \bmod 26$

E (4) $C = (4+20) \bmod 26$.

E (4) $C = (4+19) \bmod 26$.

T (19) $C = (19+18) \bmod 26$.

A (0) $C = (0+20) \bmod 26$.

T (19) $C = (19+13) \bmod 26$.

N (13) $C = (13+18) \bmod 26$.

O (14) $C = (14+20) \bmod 26$.

O (14) $C = (14+13) \bmod 26$.

N (13) $C = (13+18) \bmod 26$.

$30 \bmod 26 = 4$ (E)

$24 \bmod 26 = 24$ (Y)

$17 \bmod 26 = 17$ (R)

$37 \bmod 26 = 11$ (L)

$20 \bmod 26 = 20$ (U)

$32 \bmod 26 = 6$ (G)

$31 \bmod 26 = 5$ (F)

$34 \bmod 26 = 8$ (I)

$27 \bmod 26 = 1$ (B)

$31 \bmod 26 = 5$ (F)

2(ii) Decryption formula: $C = (P - k) \bmod 26$.

EYIRLUGIFIBF.

E(4)	$P = (4 - 18) \bmod 26$	12(M).
Y(24)	$P = (24 - 20) \bmod 26$	4(E).
R(17)	$P = (17 - 13) \bmod 26$	19(T).
L(11)	$P = (11 - 18) \bmod 26$	0(A).
U(20)	$P = (20 - 20) \bmod 26$	19(T).
G(6)	$P = (6 - 13) \bmod 26$	13(N).
F(5)	$P = (5 - 18) \bmod 26$	14(O)
I(8)	$P = (8 - 20) \bmod 26$	14(O)
B(1)	$P = (1 - 13) \bmod 26$	13(N)
F(5)	$P = (18 - 5) \bmod 26$.	

∴ By using repeating key

The Encrypted word is: EYRLUGIFIBF.

Decrypted word is: MEETATNOON.

3) Plain text : MATH.

Hill cipher : 2x2.

key k = $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$

Numerical : A=0, B=1 ... Z=25

Solu: A=0, B=1, ... I=25

Letter : M A T H.

12 0 19 7.

Block 1 : $\begin{bmatrix} 12 \\ 0 \end{bmatrix}$

Block 2 = $\begin{bmatrix} 19 \\ 7 \end{bmatrix}$

To encrypt : C = (k x p) mod 26.

~~Block I~~: $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} = \begin{bmatrix} (3 \times 12) & (2 \times 0) \\ (5 \times 12) & (7 \times 0) \end{bmatrix}$

$\neq$

~~Block II~~: $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} =$ ~~3x19 8x7~~

Block 1 : $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} = \begin{bmatrix} 36 \\ 60 \end{bmatrix} = \begin{bmatrix} 10 \\ 8 \end{bmatrix} \begin{matrix} \rightarrow K \\ \rightarrow I \end{matrix}$

Block 2 : $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix} \begin{matrix} \rightarrow T \\ \rightarrow O. \end{matrix}$

Checking whether the given key is valid or invalid

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

|det| for K = |K| = ad - bc = (3x7) - (2x5)

= 21 - 10

= 11

GCD of (11, 26) = 1.

Find matrix inverse:

$11 \times x \equiv 1 \pmod{26}$.

$11 \times 19 = 209$.

$209 \bmod 26 = 1$.

$11^{-1} = 19 \pmod{26}$

Find the inverse of $k = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$

$$\text{adj}(k) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\text{adj}(k) = \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

taking mod 26 $= \begin{bmatrix} 7 & 24 \\ 21 & 3 \end{bmatrix}$

$k^{-1} = (\det k)^{-1} \times \text{adj } k \pmod{26}$

$= 19 \begin{bmatrix} 7 & 24 \\ 21 & 3 \end{bmatrix} \bmod 26$.

$= \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$

To decrypt: $P = k^{-1} \times C \bmod 26$.

1st block $P_1 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix}$

$= \begin{bmatrix} 30 + 112 \\ 90 + 40 \end{bmatrix} = \begin{bmatrix} 142 \\ 130 \end{bmatrix}$

Taking mod 26 $= \begin{bmatrix} 142 \\ 130 \end{bmatrix} \longrightarrow 12\ (M)$

$\longrightarrow 0\ (A)$

2nd block $P_2 = \begin{bmatrix} 3 & 11 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix}$

$= \begin{bmatrix} 57+98 \\ 171+40 \end{bmatrix} = \begin{bmatrix} 155 \\ 211 \end{bmatrix}$

taking mod 26 = $\begin{bmatrix} 155 \\ 211 \end{bmatrix} \rightarrow 19\ (T)$, $\rightarrow 7\ (H)$

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4) Given:

$p=11$, $q=19$.

Public exponent $=7$.

To calculate: n, $\varphi(n)$, $\gcd(e, \varphi(n)) = 1$

$p = 11$, $q = 19$, $e = 7$, $M = 5$.

i) $n = p \times q = 11 \times 19 = 209$.

$\varphi(n) = (p-1)(q-1) = (11-1)(19-1) = 10 \times 18 = 180$

$\gcd(7, 180) = 1$.

Find key $= 7d = 1 \bmod 180$.

$180 = 7 \times 25 + 5$.

$7 = 5 \times 1 + 2$

$5 = 2 \times 2 + 1$

back substitution: $1 = 5 - 2(2)$

$= 5 - 2(7-5)$

$= 3(5) - 2(7)$.

$= 3(180 - 7 \times 25) - 2(7)$

$= 3(180) - 77(7)$

$d = -77 \bmod 180 = 103$

Key :

Public key : $(e, n) = (7, 209)$

Private key : $(d, n) = (103, 209)$

Encryption : $C = m^e \bmod n = 5^7 \bmod 209$

$5^2 = 25$, $5^4 = 625 \bmod 209$.

$= 207$.

$5^7 = 207 \times 25 \times 5 \bmod 209$.

$= 168$

Decryption : $M = C^d \bmod n = 168^{103} \bmod 209$.

$= 168^{103} \bmod 209 = 5$

5) Given :

By using classical encryption techniques (i) caesar cipher.
(ii) playfair cipher.
(iii) vigenere cipher

Solu

(i) Encryption and Decryption using caesar cipher.

assuming plain text as : MEETATNOON.

key : 7.

formula for Encryption : $C = (P + k) \bmod 26$.

Decryption : $P = (C - k) \bmod 26$

Encryption ; $C = (P + K) \bmod 26$.

M	E	E	T	A	T	N	O	O	N.
12	4	4	19	0	19	13	14	14	13.

M(12)	$C = (12+7) \bmod 26$.	19(T)
E(4)	$C = (4+7) \bmod 26$	11(L)
E(4)	$C = (4+7) \bmod 26$	11(L)
T(19)	$C = (19+7) \bmod 26$	1(B)
A(0)	$C = (0+7) \bmod 26$	7(H)
19(T)	$C = (19+7) \bmod 26$	1(B)
N(13)	$C = (13+7) \bmod 26$	20(U)
O(14)	$C = (14+7) \bmod 26$	21(V)
O(14)	$C = (14+7) \bmod 26$	21(V)
N(13)	$C = (13+7) \bmod 26$	20(U).

∴ the encrypted is TLLBHBUVVU.

Decryption ; $P = (C - K) \bmod 26$.

T	L	L	B	H	B	U	V	V	U
19	11	11	1	7	1	20	21	21	20.

T(19)	$P = (19-7) \bmod 26$.	12(M)
L(11)	$P = (11-7) \bmod 26$.	4(E)
L(11)	$P = (11-7) \bmod 26$.	4(E)
B(1)	$P = (1-7) \bmod 26$	19(T)
H(7)	$P = (7-7) \bmod 26$.	0(A)

B(1) P = (1-7) mod 26. 19 (T)

U(20) P = (20-7) mod 26. 18(N)

V(21) P = (21-7) mod 26. 14(O)

V(21) P = (21-7) mod 26. 14(O)

U(20) P = (20-7) mod 26. 13(N)

∴ The decrypted text is : MEETATNOON.

(ii) Encryption and decryption using play fair cipher.

assuming that :

plain text : MEETATNOON.

key : MONARCHY.

here I/J are considering as same matrix.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Paring up : ME - ET - AT - NO - ON.

(a) Encryption using play fair cipher

Pair	rule	Encrypted
ME	Rectangle	CL.
ET	Rectangle	KL
AT	Rectangle	RS.

NO Same row AR.

ON Same row NM

∴ the encrypted text is CHKLRS ARNM

(b) Decryption using play fair cipher

Decrypted	Rule	cipher.
CL	Rectangle.	ME
KL	Rectangle.	ET
RS	Rectangle.	AT
AR	Same row	NO
NM	Same row	ON.

∴ the Decrypted text is MEETATNOON.

(8) Encryption and Decryption using Vigenere Ciphere

(a) Encryption using Vigenere Ciphere

assuming, plaintext = MEETATNOON.

key : SUNSUNSUNS

M(12)	C = (12+18) mod 26	30 mod 26 = 4 (E)
E(4)	C = (4+20) mod 26	24 mod 26 = 24 (Y)
E(4)	C = (4+18) mod 26	17 mod 26 = 17 (R)
T(19)	C = (19+18) mod 26	37 mod 26 = 11 (L)
A(0)	C = (0+20) mod 26	20 mod 26 = 20 (U)
T(19)	C = (19+13) mod 26	32 mod 26 = 6 (G)

NO Same row AR.

ON Same row NM

∴ the encrypted text is CIKLRS ARNM

(b) Decryption using play fair cipher

Decrypted	Rule	Cipher.
CI	Rectangle.	ME
KL	Rectangle.	ET
RS	Rectangle.	AT
AR	Same row	NO
NM	Same row	ON.

∴ the decrypted text is MEETATNOON.

(8) Encryption and Decryption using Vigenere Cipher

(a) Encryption using vigenere cipher

assuming, plain text = MEETATNOON.

key : SUNSUNSUNS

M(12)	$C = (12+18) \bmod 26$	$30 \bmod 26 = 4(E)$
E(4)	$C = (4+20) \bmod 26$	$24 \bmod 26 = 24(Y)$
E(4)	$C = (4+19) \bmod 26$	$17 \bmod 26 = 17(R)$
T(19)	$C = (19+18) \bmod 26$	$37 \bmod 26 = 11(L)$
A(0)	$C = (0+20) \bmod 26$	$20 \bmod 26 = 20(U)$
T(19)	$C = (19+13) \bmod 26$	$32 \bmod 26 = 6(G)$

N(13) $C = (13+18) \bmod 26$ $31 \bmod 26 = 5$ (F).

O(14) $C = (14+20) \bmod 26$ $34 \bmod 26 = 5$ (I)

O(14) $C = (14+13) \bmod 26$ $27 \bmod 26 = 1$ (B)

N(13) $C = (13+18) \bmod 26$ $31 \bmod 26 = 5$ (F)

∴ The encrypted text is EVRLUGIFIBF.

(b) Decryption using Vigenere Cipher

E(4) $P = (4-18) \bmod 26$ 12(M)

V(21) $P = (21-20) \bmod 26$ 1(E)

R(17) $P = (17-13) \bmod 26$ 4(E)

L(11) $P = (11-18) \bmod 26$ 19(T)

U(20) $P = (20-20) \bmod 26$ 0(A)

G(6) $P = (6-13) \bmod 26$ 19(T)

F(5) $P = (5-13) \bmod 26$ 18(N)

I(8) $P = (8-20) \bmod 26$ 14(O)

B(1) $P = (1-13) \bmod 26$ 14(O)

F(5) $P = (18-5) \bmod 26$ 13(N)

∴ The decrypted texts is MEETATNOON

3) A Comparision :

200 hospital clinics transfer 1GB MRI files to a Control server

key scalability :

for 200 clinics keys : $\frac{n(n-1)}{2}$

$= \frac{200 \times 199}{2}$

$= 19,900$ keys.

RSA :

* Each clinic has one public/private key pair
* the server has one public - private key.

Total 201 keys.

AES is suitable for large files, whereas RSA is suitable for secure key exchange.

* hybrid encryption combines AES and RSA.